Project summary

Revised 5 October 2026

Item	Requirement
Required files	Q1.ipynb, Q1.pdf, Q2.ipynb, Q2.pdf
Submission	Moodle; only the last submission will be used.
Deadline	9pm Friday, Week 8 (Sydney time)
Marks	25 marks; 25% of the total course mark.
Late penalty	5% of the maximum assessment mark is deducted for each additional day (24 h) after the specified submission time; no submission is accepted 5 days (120 h) after the deadline
AI use	The use of AI tools, including but not limited to ChatGPT, GitHub Copilot, Gemini, Claude, or any other code/text generation tools, is strictly prohibited for this project. We will run detection checks. Any detected use of AI will be handled according to the UNSW academic integrity policies

Questions

Q1 (10 marks): earliest-connection queries on an edge-insertion stream.
Q2 (15 marks): an index for connected k-core and core-connectivity queries.

How to work and test

Download the supplied **Q1.ipynb and Q2.ipynb** templates from the project materials. Work on your own copy in Jupyter, JupyterLab or VS Code, or upload the notebooks to Google Colab and save a copy in Drive. A Colab link may also be supplied on the course page.

Implement the TODOs, keeping the public signatures and tests unchanged. Run cells in order from top to bottom. Local tests use deterministic fixtures and independent small-case checks; scale tests compare complete output digests. Before submission, restart the session and run all cells again. Completed notebooks must finish without exceptions.

Submission guide

Download the completed notebooks as Q1.ipynb and Q2.ipynb. Write the supporting analyses as Q1.pdf and Q2.pdf. Upload all four files to Moodle. Read each question's requirements for time and space analysis.

Revision note: Q2 explicitly requires ascending vertex IDs and supports k = 0. The templates' test fixtures have been updated; use the templates distributed with this revision. Deadline, assessment weight, late penalty and AI-use policy are unchanged.

END OF PROJECT